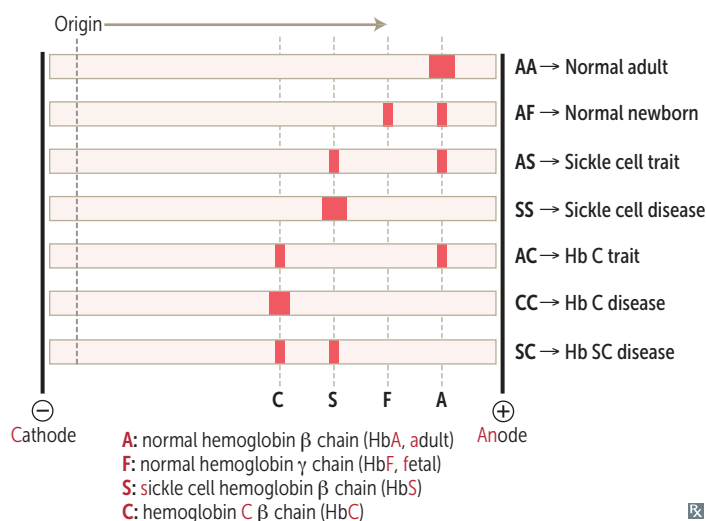


► HEMATOLOGY AND ONCOLOGY—PHYSIOLOGY

Hemoglobin electrophoresis



During gel electrophoresis, hemoglobin migrates from the negatively charged cathode to the positively charged anode. HbA migrates the farthest, followed by HbF, HbS, and HbC. This is because the missense mutations in HbS and HbC replace glutamic acid $\ominus$ with valine (neutral) and lysine $\oplus$, respectively, making HbC and HbS more positively charged than HbA.

A Fat **S**anta **C**laus can't go far.

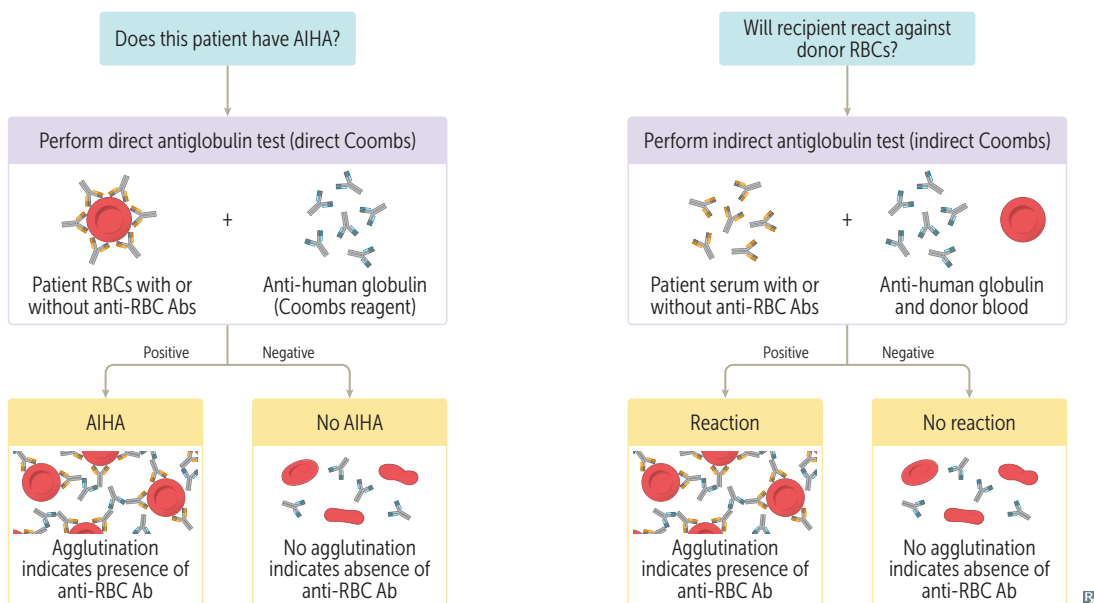
HbC is closest to the **C**athode. HbA is closest to the **A**node.

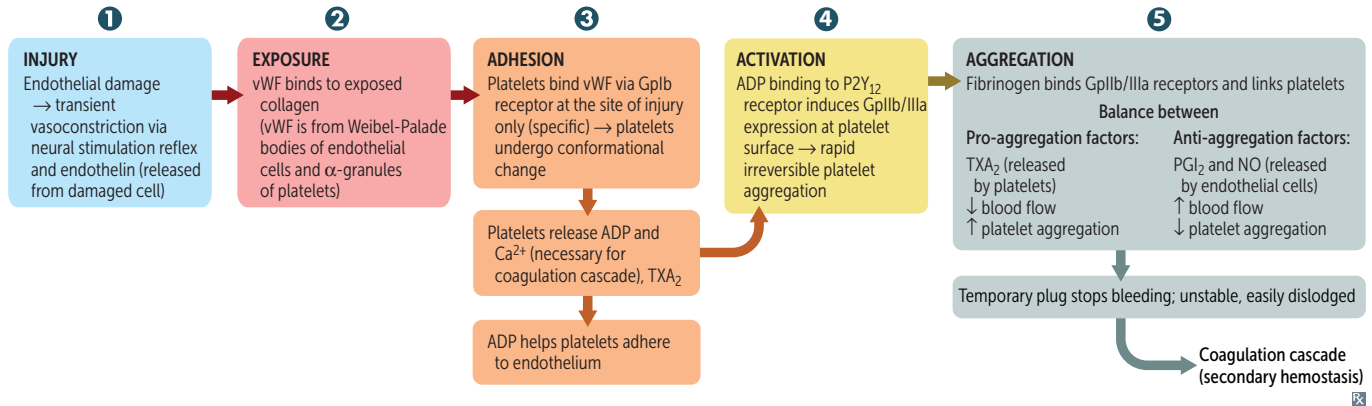
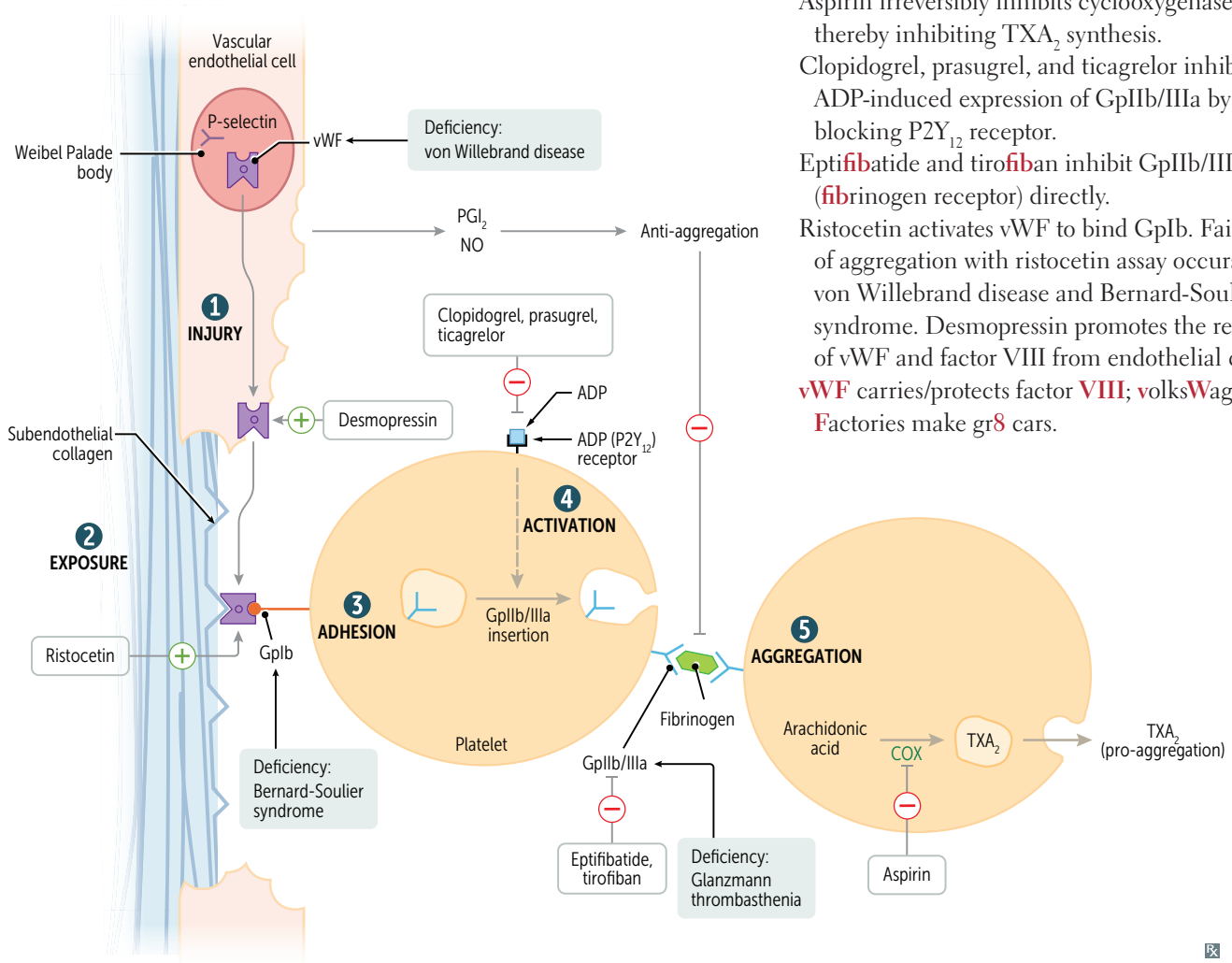
Antiglobulin test

Also called Coombs test. Detects the presence of antibodies against circulating RBCs.

Direct antiglobulin test—anti-human globulin (Coombs reagent) added to patient's RBCs. RBCs agglutinate if RBCs are (**directly**) coated with anti-RBC Abs. Used for AIHA diagnosis.

Indirect (not direct) antiglobulin test—normal RBCs added to patient's serum. If serum has anti-RBC Abs, RBCs agglutinate when Coombs reagent is added. Used for pretransfusion testing.



Platelet plug formation (primary hemostasis)**Thrombogenesis**

Formation of insoluble fibrin mesh.

Aspirin irreversibly inhibits cyclooxygenase, thereby inhibiting TXA_2 synthesis.

Clopidogrel, prasugrel, and ticagrelor inhibit ADP-induced expression of GpIIb/IIIa by blocking P2Y_{12} receptor.

Eptifibatide and tirofiban inhibit GpIIb/IIIa (fibrinogen receptor) directly.

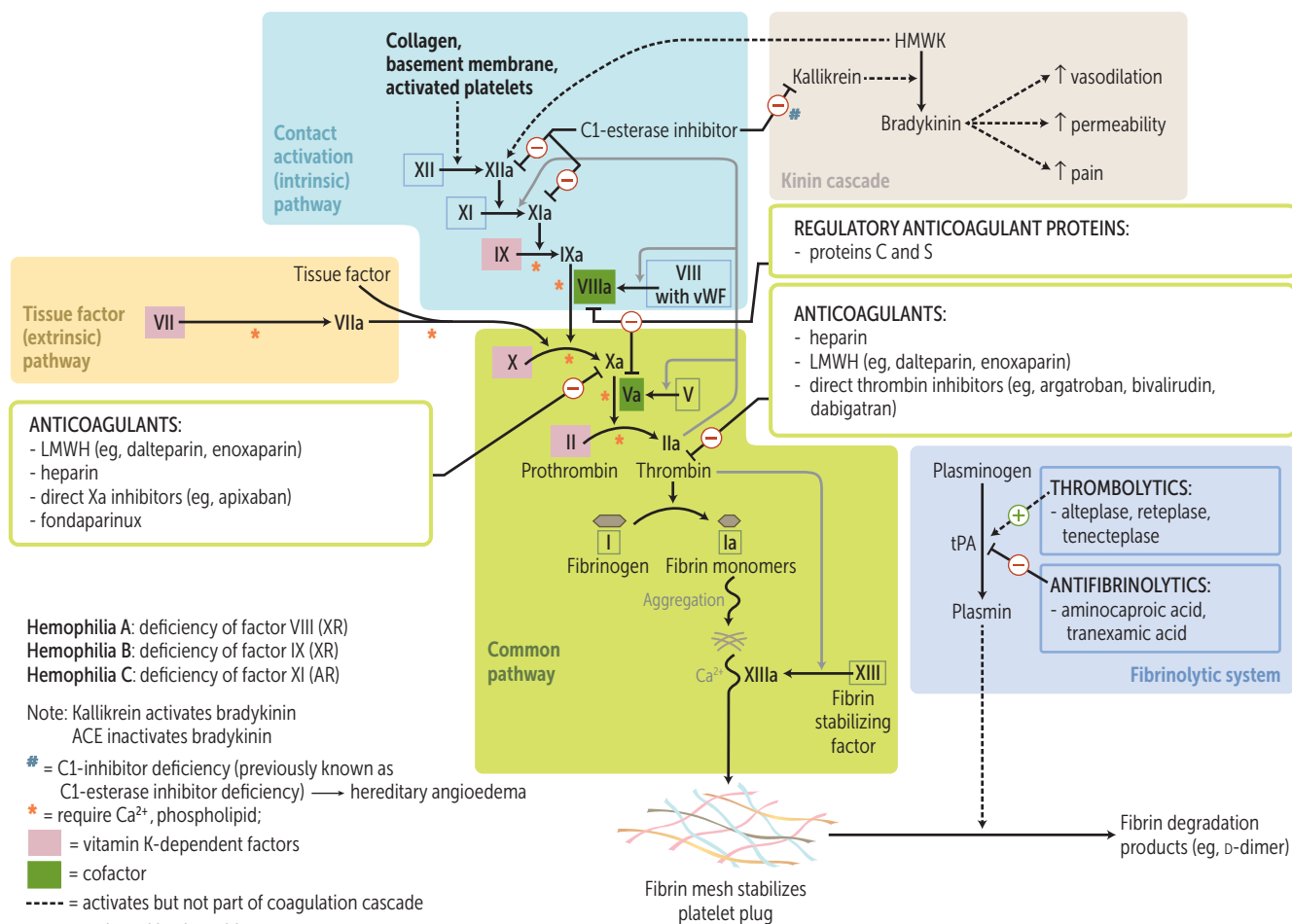
Ristocetin activates vWF to bind GpIb. Failure of aggregation with ristocetin assay occurs in von Willebrand disease and Bernard-Soulier syndrome. Desmopressin promotes the release of vWF and factor VIII from endothelial cells.

vWF carries/protects factor VIII; volksWagen Factories make gr8 cars.

Coagulation and kinin pathways

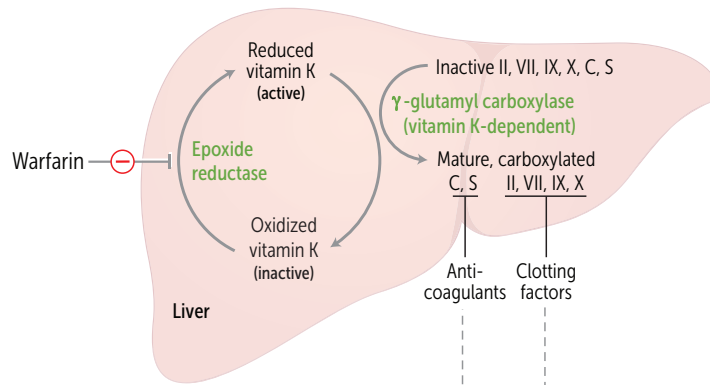
PT monitors extrinsic and common pathway, reflecting activity of factors I, II, V, VII, and X.

PTT monitors intrinsic and common pathway, reflecting activity of all factors except VII and XIII.

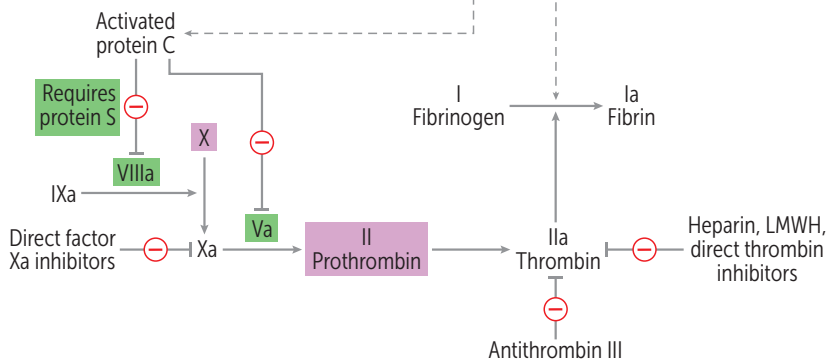


Vitamin K–dependent coagulation

Procoagulation



Anticoagulation



= vitamin K-dependent factors - - - = activates but not part of coagulation cascade
 = cofactor LMWH = low-molecular-weight heparin

Rx

Vitamin K deficiency—↓ carboxylation and maturation of factors II, VII, IX, X, protein C, protein S.

Warfarin inhibits vitamin K epoxide reductase. Vitamin K administration can potentially reverse inhibitory effect of warfarin on clotting factor synthesis (delayed). FFP or PCC administration reverses action of warfarin immediately and can be given with vitamin K in cases of severe bleeding.

Neonates lack enteric baKteria, which produce vitamin K. Early administration of vitamin K overcomes neonatal deficiency/coagulopathy. Suppression of gut flora by broad spectrum antibiotiKs can also contribute to deficiency.

Factor VII (seven)—shortest half-life.

Factor II (two)—longest (too long) half-life.

Antithrombin inhibits thrombin (factor IIa) and factors VIIa, IXa, Xa, XIa, XIIa.

Heparin enhances the activity of antithrombin. Principal targets of antithrombin: thrombin and factor Xa.

Factor V Leiden mutation produces a factor V resistant to inhibition by activated protein C. tPA is used clinically as a thrombolytic.